INCH-POUND
MIL-DTL-24643/27F
1 October 2009
SUPERSEDING
MIL-DTL-24643/27E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, 600 VOLTS, TYPE LS2AU (INCLUDING VARIATION LS2AUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS2AU and LS2AUS (double overall shield).

Construction, Non-Watertight

First	-	Copper conductor, tin-coated, ASTM B286, Size 22 – 7strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Minimum average wall thickness of 0.024 inch; telephone identification applied by Method 3.
Fourth	-	Two conductors cabled together with a lay not greater than 1½ inches.
Fifth	-	Forty two pairs cabled with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm well-rounded assembly. Cabling sequence shall be consecutive, starting with no. 1. in the center outward.
Sixth	-	An optional binder.
Seventh	-	Braided shield of tin-coated copper. Minimum coverage of 85 percent.

Construction, Variant for Type LS2AU

Eighth	-	An optional binder.
Ninth	-	Cross-linked polyolefin jacket. Minimum average wall thickness 0.060 inch.

Construction, Variant for Type LS2AUS

Eighth	-	Spirally wrapped 0.002-inch thick polyethylene (polyester film) separator tape with 25 percent minimum lap.
Ninth	-	Braided shield of tin-coated copper. Minimum coverage of 85 percent.
Tenth	-	An optional binder.
Eleventh	-	Cross-linked polyolefin jacket. Minimum average wall thickness 0.060 inch.

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TABLE I. Details.

Military part no. M24643/27	Type and size	Number of pairs	Diameter over shielded core nom. (inch)	Overall diameter	
				min. (inch)	max. (inch)
-01UO	LS2AU-40	40	1.089	1.320	1.370
-01UD	LS2AUS-40	40	1.120	1.380	1.430

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	17.71
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2500
Overall shield to overall shield (Type LS2AUS only)	200
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	5000
Overall shield to overall shield (Type LS2AUS only)	100
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Capacitance, at 1 megahertz (MHz) (pf/ft max.)	
Mutual capacitance	20

Group B:

Cold bending cable at 0±2 °C, (16-inch mandrel diameter)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

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Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Shield (conformance to material, construction and coverage)	No failure
Surface transfer impedance (Type LS2AUS only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2

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Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No Failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS2AU amd LS2AUS	1000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-031)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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